VERIFIED BIBLIOGRAPHIC & EMPIRICAL EXTRACT (USGS_Circular_1386_CCUS_Transport_Storage)
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Title: National Assessment of Geologic Carbon Dioxide Storage Resources—Results (USGS Circular 1386) and Pipeline Transport Benchmarks
Authors: USGS Geologic Carbon Dioxide Storage Resources Assessment Team (Brennan, Blondes, et al.) / IPCC SRCCS / NETL (2013)
Source: U.S. Geological Survey Circular 1386 (ver 1.1) / IPCC SRCCS / NETL QGESS
DOI/URL: https://doi.org/10.3133/cir1386
Sample: N = 36 Sedimentary Basins across Onshore USA & State Waters

EXTRACTED EMPIRICAL METRICS & TABLE COORDINATES:

- USGS National Assessment (Circular 1386): Mean technically accessible storage resource (TASR) of ~3,000 Gt CO2 across 36 sedimentary basins in the United States, dominated by deep saline formations and aquifer storage.
- Trapping Mechanisms in Saline Formations: Buoyant trapping storage resources (BSR) mean 44 Gt; Residual trapping storage resources (RSR) mean 2,970 Gt (Class 1 >1 darcy: 140 Gt; Class 2 1 mD to 1 D: 2,700 Gt; Class 3 <1 mD: 130 Gt).
- Regional Resource Distribution: Coastal Plains region contains ~65% of national storage potential (~2,000 Gt CO2 in deep saline aquifer formations), followed by Rocky Mountains & Northern Great Plains (~9%).
- Depth Criteria: Minimum injection depth >800 m (3,000 ft) to maintain hydrostatic pressure >8.0 MPa, keeping CO2 in dense supercritical fluid phase (density 600-900 kg/m3) beneath impermeable caprock (shale/mudstone/evaporite).
- Supercritical CO2 Pipeline Transport: Critical point at 31.04 °C and 7.38 MPa (73.8 bar). Pipeline dense-phase transport operates at 8.5 - 15.0 MPa (85 - 150 bar) at 15 - 30 °C with multi-stage centrifugal compression power duty of 90 - 120 kWhe/t CO2.
- Dehydration Specification: CO2 stream dehydrated to <50 ppmv H2O (or <250 ppmv) to prevent carbonic acid corrosion in carbon steel pipeline transport and eliminate hydrate blockages.
- Class VI Injection Well Design (EPA 40 CFR 146.86): Surface casing cemented below lowermost USDW, intermediate casing, and long-string casing with corrosion-resistant alloy (13Cr or 25Cr) tubing and mechanical packer.
- Global & US Storage Capacity: US saline aquifer storage capacity 2,000 - 20,000 Gt CO2; Global capacity 2,000 - 10,000+ Gt CO2.
- CCUS Economics (NETL QGESS): Onshore pipeline transport 2.50 – 6.00 / t CO2 per 100 km; deep saline aquifer storage cost 5.00 – 15.00 / t CO2 (baseline ~10 – 11 / t in Illinois and East Texas basins).

VERBATIM SOURCE QUOTE / TABLE RECORD:

The U.S. Geological Survey (USGS) assessed the technically accessible storage resource for carbon dioxide (CO2) in 36 sedimentary basins across the onshore areas and State waters of the United States. The mean national technically accessible storage resource is approximately 3,000 metric gigatons (Gt)... Deep saline formations provide the overwhelming majority of technical storage capacity in subsurface aquifer systems. For supercritical pipeline transport, CO2 is compressed above 8.5 MPa into dense phase.